Cyclone

Competition review and gap report

Reviewed 10 November 2026 | Meta VR Start Developer Competition 2026

Recommendation: keep the concept and close the evidence and usability gaps. Cyclone distinctively combines spatial rehearsal, hand-based amendment and signed decisions. Its strongest moment is the robot-arm proposal: reveal a breach, change 250 to 180 degrees/second, then deliberately commit. The build is credible, but its evidence and finish need work.

Invest this week in making that moment reliable and understandable, before adding scenarios, multiplayer or a conversational assistant.

Planning scorecard. These provisional reviewer estimates assume eligibility and a complete entry. They are not official scores or predictions. Each official category carries 25%; the target is an ambition contingent on passing this report's checks. [S3]

Criterion	Present estimate	Working target	Main reason for the gap
Innovation & Creativity	19 / 25	22 / 25	Strong combination; the headset advantage and adaptive agent behavior need evidence.
Experience Design	15 / 25	21 / 25	Settings overflow, incomplete onboarding recovery, and unproven reach in real rooms.
Technical Implementation	15 / 25	21 / 25	Good test foundation; decision-proof association and execution gating need correction; hardware performance is unknown.
Polish & Presentation	13 / 25	21 / 25	Functional miniature, but unfinished hierarchy, no verified final video, and overstated submission claims.
Total	**62 / 100**	**85 / 100**	Improve proof, clarity and finish before expanding scope.

What already deserves credit

- The demo's rehearsal, gestures, local signing, drills and persistence work independently of the external policy service.
- Pull, twist, hold and push have understandable purposes.
- Risk uses shape, pattern, motion and sound alongside color; one-handed operation and captions have foundations.
- The build and 96 automated tests pass. The public demo enters an immersive browser session.

What changes the assessment

Settings buttons measure 80 cm wide inside a 22 cm panel. A mocked unrelated dossier can be labeled verified; a timed-out drill earns a 100% catch score; and model changes alter historical replay. These weaken Cyclone's interaction and trust story.

Scope: source, tests, saved evidence, public browser checks and competition material. Repository: /Users/bonus/develop/products/Cyclone, HEAD 481fa6d. Source unchanged. Hardware, real-room behavior and production integration remain unverified.

Entry readiness and award strategy

The deadline is **18 November 2026, 21:00 Stockholm time** (12:00 PST). The submitted build must remain accessible and unchanged after the deadline through the required judging availability period; the announced winner date is approximately 11 December. [S1, S2]

Entry gate	Assessment	Required action
Track and division	Productivity / New Experience is defensible.	Position it around a supervisor's recurring review session.
Originality window	Visible Git history begins 28 October.	Owner must confirm the project was also conceived within the eligible window beginning 24 September; Git dates alone do not establish that.
Membership and registration	Not established by repository evidence.	Confirm Start membership, developer access, registration and entrant/team eligibility.
Judge access	Public WebXR URL launches.	Recheck from a clean headset browser and retain an immutable release.
Video	Shot list exists; final public video is not evidenced.	Supply a public YouTube/Vimeo video under three minutes. Authentic emulator footage is allowed; the repository's headset-only H4 dependency is stricter.
Submission text	Draft exists, but claims need correction.	Confirm launch date and align every assertion with observed behavior.

Eligibility, rights, applicable policies and absence of disqualifying sponsorship/funding remain owner confirmations. Nothing in this review establishes those account or ownership facts. Official rules govern entry; this is a readiness assessment. [S2]

Award order

Award	Fit today	How to strengthen the case
Best New Productivity Experience ($70,000)	Best primary track; value remains a hypothesis.	Show fewer wrong decisions or faster accurate review for a defined supervisor.
Boldest Original Concept ($25,000)	Strong conceptual fit.	Demonstrate how depth, hands and the physical review surface materially help the core task.
Best Agentic Interaction ($25,000)	Promising, but conditional.	Show an agent adapting a proposal to a changed goal or constraint. Scripted playback alone is weak evidence of AI being central.
Best First Five Minutes ($25,000)	Plausible after usability work.	Complete unfamiliar-user runs with a clear payoff and no coaching.
Reason to Come Back / Accessibility Forward	Secondary opportunities.	Establish varied useful sessions and genuinely usable access paths before emphasizing either.

These are alternative award opportunities, not additive points: the rules limit an entry to one prize. Multiplayer is a poor last-week investment for this build. [S1, S2]

The current project gallery is unpublished. I therefore cannot rank Cyclone against the 2026 submitted field or estimate a win probability. The benchmarks below are established adjacent products and prior competition examples. [S4]

Innovation, uniqueness and competitive position

Cyclone's differentiator is the combined decision workflow. Neither digital twins, approval queues, mixed reality nor signatures is independently novel. The unusual proposition is letting a person understand a spatial consequence, physically alter the proposed action and preserve the basis of the decision in one short session.

Adjacent benchmark	Existing overlap	Where Cyclone can differentiate
LangChain / LangGraph human review [S5]	Agents can pause for approval, editing or rejection of tool calls.	Turn physical consequences into an inspectable rehearsal before approval.
NVIDIA Omniverse [S6]	Industrial digital twins, robotics simulation and operational change testing.	A lightweight human review interface for individual agent proposals, with fast headset access.
ABB RobotStudio HyperReality [S7]	Industrial robot simulation, virtual commissioning and AI-related simulation workflows.	Concentrate on human decision quality and accountability; do not compete on industrial simulation fidelity.
ShapesXR [S8]	Spatial editing and synchronized mixed-reality collaboration using shared anchors.	Make hand interaction serve authorization and decision history, rather than general design collaboration.
Previous competition winners [S9]	Examples such as Inkphony, DreamSpace and Chores.gg connect an obvious activity to spatial interaction or the room.	Make Cyclone's value equally immediate: one proposal, one visible problem, one meaningful correction.

These comparisons do not establish that no similar product exists. Avoid a "world first" claim. A defensible claim is that Cyclone combines these mechanisms in a distinctive hands-first review experience.

Why a headset is worth using

The best case is an operations supervisor or robotics instructor reviewing motion, clearances and dependent machinery during a short handover or training session. Depth and embodied comparison may help them understand consequences that are awkward to infer from parameters.

The weak case is ordinary approval work that a desktop can handle in seconds. A tabletop scene can also be rendered on a laptop, and signed records are platform-independent. Cryptography alone therefore does not justify VR. The claim that this concept could not exist on another platform is currently too strong.

Test the useful claim instead: does Cyclone improve accurate review enough to justify wearing the headset? Use a desktop comparison with equivalent information and a usable 3D view, rather than an intentionally poor JSON-only screen. Measure decision time, unsafe approvals, unnecessary vetoes and requests for help. A small pilot can reveal usability problems; it cannot establish industrial safety or broad productivity gains.

Positioning to use now: a five-minute spatial review desk for people supervising physical AI. Keep the factory as the concrete demonstration. Describe real execution as a future integration and retain the visible distinction between a model rehearsal, a policy authorization and an actual machine result.

Experience gaps that judges can encounter

Priority meanings: **P0** before a submission candidate; **P1** before the 17 November freeze; **P2** only after those checks pass. These are implementation priorities, not official competition severity levels.

Gap	Evidence and consequence	Closure check
G01 - P0: settings overflow	Current public runtime: root 22 cm wide; all ten buttons 80 cm wide, despite an authored 9.6 cm width. The second column is displaced far outside the intended panel.	Resolve component sizing/minimum constraints; fit the full panel; activate every control with hands and the supported gaze path.
G02 - P1: onboarding is skipped after an unfinished visit	A visit is recorded at startup, and any later visit defaults to handover. An early exit or refresh can bypass training.	Persist onboarding completion separately; offer resume/restart until the first shift is completed.
G03 - P1: access claims exceed behavior	"Eyes-only" still requires a pinch. Inspect selection uses head direction; a genuine eye-gaze selection path is not established by that code.	Name the current mode accurately; verify supported eye input and one-handed navigation, including settings and recovery.
G04 - P1: real-room fit is incomplete	Plane placement has an edge-distance fallback, but no full board-fit or all-controls reach check. Static reach tests use a nominal head and omit dynamic UI.	Test small, rotated and nearby side tables, different seated heights and no-table fallback; reposition or use the plinth when fit fails.
G05 - P1: performance evidence is desktop-only	Saved evidence reports 226 draw calls and about 29.9k triangles. Fresh build has a 7.90 MB main chunk, 2.05 MB gzipped. Neither proves headset frame rate.	Measure cold start and sustained frame timing on Quest 3/3S; meet the 60 fps criterion and aim for the app's 72 fps target.
G06 - P1: the action is harder to read than the concept	Small dense cards, similar desk fittings and temporary captions carry much of the instruction.	Give the current proposal, breach and next gesture clear hierarchy; keep rationale progressive; test captions and contrast on-device.



Current public settings overflow, cropped from a browser emulator screenshot

The screenshot is a current desktop emulator observation, not a headset capture or hands-only test. Runtime measurements confirm the overflow independently of the camera view. Device performance expectations come from the judging criteria. [S3]

Source pointers: settings.uikitml:31, settings.ts:75, app.ts:93, card.ts:76, twin.ts:137, layout.ts:67, perf.md.

Decision integrity and product-logic findings

G07 - P0 for a verified-decision claim: associate the dossier with the reviewed action. The verifier checks signature validity, coverage, trust source and dossier ID. It does not compare the dossier's signed contents to the expected intent ID/hash and amended action. In a mocked relay run, an unrelated valid fixture was marked verified even though it contained neither the held ID nor held hash. Require signature-covered evidence of the exact intended binding and decision before granting that status. This was a mock-only reproduction, not a production-service exploit. verify.ts:80, seal.ts:38.

G08 - P0 for a fail-closed claim: make verification affect the approval outcome. A second mocked run returned APPROVE, verified: false and stored state resolved. The twin's execution-animation handler checks the decision kind but not its verification. Thus the UI can animate an approval while the seal says it is unverified. Separate authorization, proof-pending and verified execution eligibility. Gate any execution signal on the correct verified binding; do not present a failed proof as completed execution. No real machine is currently actuated by this code. decide.ts:158, decide.ts:253, twin.ts:96.

G09 - P1: preserve the rehearsal that was actually reviewed. The saved reference hashes tool, asset and arguments, not the model or scene. Doubling the arm's braking-deceleration parameter changed the 250-degree/second result from a walkway breach to no breach while leaving the reference identical. Seal pickup regenerates motion using the current model. Preserve a versioned rehearsal snapshot and its digest. Keep later "what-if" replay distinct from the original record. rehearsal.ts:53, seal.ts:115.

G10 - P1: do not reward inaction as vigilance. Letting a drill expire produced REJECT / expired / timeout and the score "100% caught (1/1)". Current catch-rate scoring also lacks a measure of unnecessary vetoes. Count timeout as missed or unassessed, never caught; distinguish correct catches, misses and false alarms. This supports the productivity story as well as the habit loop. shift.ts:266, calibration.ts:13.

G11 - P1: make delegation bounded and revocable. The trust helper accepts an unoffered rule; its matcher accepted a high-risk, -100% conveyor change because it only checks agent, tool, single-approval mode and an upper limit. Rules lack explicit lower bounds, asset scope, expiry and individual revocation. Tighten those conditions and provide revoke/suspend. Three clean approvals are a progression mechanic, not evidence of statistical reliability. Live auto-approval is explicitly refused by the relay; this finding concerns demo delegation and future policy design. trust.ts:17, trust.ts:77, decide.ts:109.

Capability gap register

The repository marks **27 capabilities verified, 21 partial, one blocked and one not started**. Those are project-maintained statuses, not independent certifications. The groups below cover all 23 unresolved rows; several green rows also need reopening because of findings G01-G11.

Capability group	Open IDs	Gap and acceptance evidence
Live integration	C12, C35, C36, C39, C48, C49	No demonstrated production round trip or recorded live dossiers. Provision the policy and credentials, then capture hold, decision, matching signed proof and refusal cases.
Hash conformance	C13	Browser/reference parity tests pass, but the installed CLI reproduces 10 of 11 top-level corpus files and fails policy-projection.json. Reconcile the supported verifier/spec versions and retain the failing result until resolved.
Inspection and sensory feedback	C01, C02, C24, C28	Assert tether target, source position, full-rationale interaction and every cue/caption; conduct an audible review.
Onboarding and storage	C14, C19	Show an unfamiliar user completing the first shift, interrupted onboarding recovery, reset via the real controls and storage-denied operation.
Physical placement	C07, C20, C21, C22	Demonstrate anchoring, actual desk-edge veto, full control reach and narrow-field-of-view legibility in more than one room.
Historical replay	C11	Verify reload-and-replay and freeze the original model/scene evidence; persistence alone is insufficient.
Finish and device performance	C30, C31, C40	Correct settings geometry; finish visual hierarchy; profile the complete shift on hardware.
Submission video	C47	Produce, review and publish the final under-three-minute recording. A shot list is not a deliverable.
Common proof-bundle format	C50	Not started. Keep explicit scope and a time box; it must not displace launchability and interaction fixes.

Source: CAPABILITIES.md. In particular, green status for drill scoring, delegation, access modes and submission copy should not prevent reopening those items.

Live amendment is an implementation gap, not just a credential gap. The current amended-approval branch always ends in an unsupported response, including after its configuration flag is enabled. Use the documented counter-offer/re-proposal path until an actual amended-approval route is implemented and verified. Do not advertise the configuration flag as working functionality. decide.ts:167.

The daily-use story needs correction. The existing decision log records a five-minute live intent lifetime, which conflicts with reviewing the same actionable approvals "held overnight." Retain an overnight history if useful, then re-propose and revalidate current actions. Demo handover contains six authored proposals and a two-template optional drill pool; it demonstrates a return-session structure, not sustained retention or a real backlog. DECISIONS.md:27, handover.json, drills.json.

Features with the best scoring value

Estimates assume one developer who already knows the project. They are rough engineering effort, exclude waiting for hardware/service access, and are not promises of points. Benefits overlap; do not add imagined point gains per feature.

Rank	Feature	Why it helps	Effort and decision
1	Compare original and amended actions	Makes the spatial benefit and the user's contribution immediately visible. Supports innovation, experience and presentation.	About 8-12 hours after fixes. Best first addition.
2	Immutable rehearsal receipt	Makes the audit claim demonstrable and turns the seal into a useful object. Supports innovation and technical credibility.	About 8-12 hours for a minimal snapshot; external proof linkage may take longer.
3	Scoped, revocable delegation	The user grants limited authority and sees why interruptions decrease. Supports recurring value and control.	About 4-8 hours for a minimal rule editor/revoke path plus tests. First repair G11.
4	Goal-driven re-proposal	An agent responds to a changed constraint instead of replaying a fixed example. Strengthens the Agentic award argument.	About 12-24 hours or more. Only if the core and evidence are already complete.
5	Desk layout that adapts to available space	The actual table changes placement and reachable controls, making mixed reality more useful.	About 8-16 hours plus device testing. Basic fit/fallback repairs are already P1.

Feature 1: show the difference at the same instant. Keep the original path as a thin patterned trail and show the amended action on the same timeline. Let the user switch the active ghost; avoid doubling transparent rendering cost. In the arm example, the original sweep crosses the marked walkway and the amended sweep stays clear. Show values and model-derived clearance or duration only where the model supports them.

Acceptance: both alternatives use the same starting state and model; moving the dial updates the comparison immediately; shape/text distinguish them without color; an unfamiliar user can explain the difference; hardware frame time remains within budget. This is a small extension to an existing, strong interaction.

Feature 2: make a seal preserve its evidence. Store original and amended action bindings, model version/parameters, zone geometry, time/units and the reviewed trajectory or immutable reference. Sign or hash that packet and link it clearly to the authority record. Distinguish the client's rehearsal receipt from the policy service's signed claims; do not imply the service signed evidence it did not cover.

Acceptance: changing a model later does not change historical replay; tampering invalidates the receipt; a wrong-action dossier cannot validate it; detail shows what was simulated, verified and not established. Keep cryptographic detail behind the physical seal rather than in the primary decision card.

Build no more than two additions before freeze. Feature 1 is the clearest visual improvement; Feature 2 repairs the product's central promise. If those threaten device testing or the final recording, finish the fixes and the existing core loop first.

Final-week plan and proof to collect

Use early hardware results to shape the release. Do not leave room-fit or hand-tracking discoveries until after polishing the trailer. The 17 November freeze is the project's own buffer; the official cutoff is the following day. [S1]

Date, Stockholm	Work	Exit evidence
10 November	Correct settings sizes; audit every accessible control; fix unfinished-visit routing.	Every setting reachable; refresh/return keeps onboarding understandable.
11 November	Repair proof association and unverified completion; fix drill scoring and unsafe delegation matching.	Wrong-intent, missing-proof, timeout and rule-boundary regressions pass.
12 November	Run headset checks in two rooms and a no-table setup. Profile cold start, heavy scene and hands.	Recorded reach/FoV results and sustained Quest frame timing.
13 November	Add the original-versus-amended comparison if the baseline passes.	One short, clear arm demonstration with stable frame time.
14 November	Finish a minimal immutable receipt; tighten captions and hierarchy.	Reload/model-change/tamper checks; complete first-shift recording.
15 November	Run unfamiliar-user trials; fix the largest friction. Validate live mode if access is available.	Task outcomes, failure notes and an accurately labeled live or demo evidence set.
16 November	Final video, text, screenshots, launch-date confirmation and clean-device judge access.	Complete reviewable submission package.
17 November	Fix-only release candidate; tag and preserve hosted build.	Final smoke check and frozen artifact.
18 November	Submit early, before 21:00.	Entry receipt and working public links.

Acceptance checks worth more than another feature

- **Unfamiliar-user pilot:** start with five people, including someone representative of the intended operator/instructor. Record completion, assistance, wrong approvals and unnecessary vetoes. Suggested internal targets: first useful action within 45 seconds, first payoff within 90 seconds and a complete shift within five minutes. These are goals, not measured results.
- **Fair desktop comparison:** counterbalance equivalent scenario sets to reduce learning effects. Compare median accurate-decision time and error counts; record the sample size and limitations. Avoid publishing a productivity percentage without observed data.
- **Device checks:** controllers unpaired; both hands and the one-handed option; temporary tracking loss; narrow FoV; different desk shapes/heights; pause/resume; reload; fallback; a full five-minute session. Record device/OS/build and raw frame-time measurements.
- **Decision checks:** wrong binding, wrong dossier, amended hash mismatch, expired intent, missing/untrusted proof and rule revocation all behave as documented. Separate permission to act from evidence that a machine acted.

If time slips, remove optional additions from this week's plan, not from the capability register. Preserve the existing scope explicitly and record remaining dependencies. Do not add multiplayer, real PLC control, full voice navigation, new scenario packs or decorative AI chat during the freeze window.

Submission story and demonstration

Suggested positioning: Cyclone is a hands-first review desk for people supervising physical AI. Rehearse a proposed machine action, reshape it with your hands, and keep a record of the decision. State plainly that the competition demo runs a miniature simulation.

Keep the conveyor tutorial in the app; lead the trailer with the arm-and-walkway consequence. Explain the useful action before the architecture.

Time	Real app footage	Point to communicate
0:00-0:08	Arm proposal lands beside the marked walkway.	An agent wants to act; a person must understand the consequence.
0:08-0:35	Inspect and scrub to the breach.	Spatial rehearsal exposes what a parameter alone hides.
0:35-0:55	Twist 250 to 180; compare paths if built.	The person changes the proposal and sees the difference.
0:55-1:15	Release a hold to abort, then deliberately complete it.	The commitment gesture has a clear consequence.
1:15-1:40	Pick up the seal; show original replay and honest verification scope.	The decision has a useful, understandable record.
1:40-2:00	Locked oven, veto, then drill feedback.	Boundaries are visible and errors teach something.
2:00-2:25	Handover and bounded/revocable rule if completed.	There is a practical reason to return.
2:25-2:40	One-handed/contrast view and quiet end card.	Clear access, platform fit and demo/live labeling.

Only show additions after they work. Use an actual emulator recording if headset footage is unavailable; keep its status clear and remove development overlays from the finished capture where supported. Do not replace missing product behavior with generated footage. [S1, S3]

Current draft claim	Safer, accurate direction
"No buttons, no menus"	Describe gesture-based decisions; settings do use a panel and buttons.
"Eyes-only"	Say gaze-plus-pinch unless a no-hand navigation path is actually implemented.
Ready for VR Glasses / fully inside their FoV	Say configured for the target; specify what device and FoV checks were actually completed.
Review approvals held overnight	Explain demo handover; live actions need fresh, unexpired proposals.
Live approvals and amendments work	Describe the exact demonstrated route; use counter-offers for the unsupported amended path.
Signed record preserves exactly what was rehearsed	Claim exact historical replay only after versioned evidence is preserved and linked correctly.

Confirm the proposed Q1 2027 launch date. Sources: SUBMISSION.md, VIDEO.md.

Verification record and sources

Evidence	Result and scope
App typecheck, tests and production build	Passed; 50 tests. Build warnings include a large JavaScript chunk and dependency annotations.
Relay typecheck and tests	Passed; 46 tests using fixtures/mocks.
Agent typecheck and dry run	Passed; four requests constructed; nothing sent to a live service.
Separate verifier CLI check	Exit 1: 10 top-level vector files reproduced, policy-projection.json failed. Passing unit parity is not a full CLI conformance pass.
Public demo smoke check	Loaded the public URL and entered an immersive session; plinth fallback and first token observed; no startup console errors observed.
Current UI probe	Ten 80 cm buttons inside a 22 cm root; screenshot and computed layout agree. Controls were inspected programmatically, not proven usable by hands.
Focused logic probes	Reproduced timeout-as-catch, overly broad rule matching, model-dependent replay with unchanged reference, unverified resolved approval and unrelated-dossier verification in mocks.
Saved interaction evidence	Repository reports 15/15 onboarding and 11/11 extra emulator checks, dated 28 October. Those full hand suites were not rerun in this review.

The local IWSDK managed-browser launcher could not install its Chromium build because its download connection failed. This is a review-environment limitation, not an observed Cyclone defect. Current public checks used a separate working browser tool. No real headset, production policy workspace or actuator was used; no live-service correctness or physical safety claim follows from these tests.

Official and competitive sources, accessed 10 November 2026

Ref	Source	Used for
S1	Competition overview	Deadline, submission media, hosting and award categories.
S2	Official rules	Eligibility, originality, freeze/availability and prize limit.
S3	Judging criteria and process	Equal weighting, experience quality, hardware performance and honest presentation.
S4	2026 project gallery	Gallery not yet published; current-field ranking unavailable.
S5	LangChain human-in-the-loop documentation	Existing approval/edit/reject workflow.
S6	NVIDIA Omniverse	Simulation and industrial digital-twin adjacency.
S7	ABB RobotStudio HyperReality	Robot simulation and virtual-commissioning adjacency.
S8	ShapesXR shared anchors	Existing mixed-reality collaboration and spatial alignment.
S9	Meta's previous competition winners	Examples of clear activity/platform fit; not the 2026 field.